

Package ‘tlsR’

April 17, 2026

Type Package

Title Detection and Spatial Analysis of Tertiary Lymphoid Structures

Version 0.3.0

Date 2026-04-02

Description Fast, reproducible detection and quantitative analysis of tertiary lymphoid structures (TLS) in multiplexed tissue imaging. Implements Independent Component Analysis Trace (ICAT) index, local Ripley's K scanning, automated K Nearest Neighbor (KNN)-based TLS detection, and T-cell clusters identification as described in Amiryousefi et al. (2025) <[doi:10.1101/2025.09.21.677465](https://doi.org/10.1101/2025.09.21.677465)>.

License MIT + file LICENSE

URL <https://github.com/labsyspharm/tlsR>

Depends R (>= 4.0.0)

Imports dbscan (>= 1.1-10),
fastICA (>= 1.2-3),
FNN (>= 1.1.3),
spatstat.explore (>= 3.0-0),
spatstat.geom (>= 3.0-0),
ggplot2 (>= 3.4.0),
rlang (>= 1.0.0),
grDevices,
graphics,
stats,
methods

Suggests knitr,
rmarkdown,
testthat (>= 3.0.0)

Encoding UTF-8

LazyData true

RoxygenNote 7.3.3

VignetteBuilder knitr

Config/testthat/edition 3

Contents

tlsR-package	2
calc_icat	3
detect_tic	4
detect_TLS	5
plot_TLS	6
scan_clustering	7
summarize_TLS	9
toy_ldata	10

 tlsR-package

tlsR: Detection and Spatial Analysis of Tertiary Lymphoid Structures

Description

Fast, reproducible detection and quantitative analysis of tertiary lymphoid structures (TLS) in multiplexed tissue imaging data.

Typical workflow

1. Load or prepare a named list of data frames (`ldata`), one per tissue sample. Each data frame must contain columns `x`, `y` (spatial coordinates in microns), and phenotype (character: "B cell" / "T cell" / other).
2. Run `detect_TLS` to label B+T co-localised regions.
3. (Optional) Run `scan_clustering` to identify windows of significant immune clustering via local Ripley's L.
4. Run `calc_icat` to score the internal linearity/organisation of each detected TLS.
5. Run `detect_tic` to identify T-cell clusters outside TLS.
6. Use `summarize_TLS` to obtain a tidy summary table.
7. Use `plot_TLS` to produce publication-ready spatial plots.

Author(s)

Maintainer: Ali Amiryousefi <ali_amiryousefi@hms.harvard.edu> ([ORCID](#))

Authors:

- Jeremiah Wala <jeremiah_wala@dfci.harvard.edu> ([ORCID](#))

Other contributors:

- Peter Sorger ([ORCID](#)) [contributor]

References

Amiryousefi et al. (2025) [doi:10.1101/2025.09.21.677465](https://doi.org/10.1101/2025.09.21.677465)

See Also

Useful links:

- <https://github.com/labsyspharm/tlsR>

calc_icat

*Calculate ICAT (Immune Cell Arrangement Trace) Index***Description**

Quantifies the spatial spread and linear organisation of cells within a detected TLS. FastICA is applied to the (x, y) coordinates of TLS cells to estimate independent components; the mixing matrix is used to reconstruct the data, and the ICAT index is defined as the normalised trace-standard-deviation of the reconstructed coordinates.

The index is always non-negative because it measures the average spatial spread per cell rather than the signed trace of the mixing matrix (which can be negative due to ICA sign ambiguity). Higher values indicate a more spatially extended, structured cluster.

Usage

```
calc_icat(patientID, tlsID, ldata = NULL)
```

Arguments

patientID	Character. Sample name in ldata.
tlsID	Numeric or integer. TLS identifier (value of tls_id_knn for the TLS of interest).
ldata	Named list of data frames, or NULL to use the global ldata object (deprecated; pass explicitly).

Details

The ICAT index is computed as follows:

1. Centre the (x, y) coordinates of TLS cells.
2. Run fastICA with 2 components.
3. Reconstruct $\hat{X} = SA^T + \mu$.
4. Let v_1, v_2 be the marginal variances of $\hat{X}$.
- 5.

$$\text{ICAT} = 100 \times \frac{\sqrt{v_1 + v_2 + 2\sqrt{v_1 v_2}}}{\text{nrow}(X)}$$

If the requested TLS contains fewer than 3 cells, or FastICA does not converge, the function returns NA_real_ with an informative message rather than throwing an error.

Value

A single non-negative numeric value (the ICAT index), or NA_real_ if computation is not possible (fewer than 3 cells, or FastICA did not converge).

Examples

```
data(toy_ldata)
ldata <- detect_TLS("ToySample", k = 30, ldata = toy_ldata)
if (max(ldata[["ToySample"]]$tls_id_knn, na.rm = TRUE) > 0) {
  icat <- calc_icat("ToySample", tlsID = 1, ldata = ldata)
  print(icat)
}
```

detect_tic

Detect Tumor-Infiltrating T-cell Clusters (TIC)

Description

Applies HDBSCAN to T cells that lie outside of previously detected TLS regions to identify spatially compact T-cell clusters (TIC). Phenotype labels "T cell" and "T cells" are both accepted.

Usage

```
detect_tic(sample, min_pts = 10L, min_cluster_size = 10L, ldata = NULL)
```

Arguments

sample	Character. Sample name in ldata.
min_pts	Integer. HDBSCAN minPts parameter: minimum cluster size (default 10). Smaller values detect more, smaller clusters.
min_cluster_size	Integer. Minimum number of T cells for a HDBSCAN cluster to be retained; smaller clusters are merged back into noise (label 0). Default 10.
ldata	Named list of data frames, or NULL to use the global ldata object (deprecated; pass explicitly).

Value

The input ldata list with the sample data frame augmented by one new column:

tcell_cluster_hdbscan Integer. 0 = noise / not a T-cell cluster; positive integer = TIC cluster ID. Non-T-cell rows receive NA.

Examples

```
data(toy_ldata)
ldata <- detect_TLS("ToySample", k = 30, ldata = toy_ldata)
ldata <- detect_tic("ToySample", ldata = ldata)
table(ldata[["ToySample"]]$tcell_cluster_hdbscan, useNA = "ifany")
```

detect_TLS

*Detect Tertiary Lymphoid Structures using a KNN-density approach***Description**

Identifies TLS candidates by finding regions of high local B-cell density that also contain a sufficient number of nearby T cells (B+T co-localisation). Phenotype labels "B cell" and "B cells" (and their T-cell equivalents) are both accepted.

Usage

```
detect_TLS(
  LSP,
  k = 30L,
  bcell_density_threshold = 10,
  min_B_cells = 50L,
  min_T_cells_nearby = 10L,
  max_distance_T = 50,
  ldata = NULL
)
```

Arguments

LSP	Character. Sample name in ldata.
k	Integer. Number of nearest neighbours used for density estimation (default 30, calibrated for 0.325 um/px imaging).
bcell_density_threshold	Numeric. Minimum average 1/k-distance (in microns) for a B cell to be considered locally dense (default 15).
min_B_cells	Integer. Minimum B cells per candidate TLS cluster (default 50).
min_T_cells_nearby	Integer. Minimum T cells within max_distance_T microns of the candidate cluster centre (default 30).
max_distance_T	Numeric. Search radius (microns) for T-cell proximity check (default 50).
expand_distance	Integer. The extended values from the boundary of the detected B-cells clusters that the T cells are being integrated (default 80).
ldata	Named list of data frames, or NULL to use the global ldata object (deprecated; pass explicitly).

Value

The similarly formatted ldata list, with the data frame for LSP augmented by three new columns:

tls_id_knn Integer. 0 = non-TLS cell; positive integer = TLS cluster ID.
 tls_center_x Numeric. X coordinate of the TLS centre for TLS cells; NA otherwise.
 tls_center_y Numeric. Y coordinate of the TLS centre for TLS cells; NA otherwise.

Examples

```
data(toy_ldata)
ldata <- detect_TLS("ToySample", k = 30, ldata = toy_ldata)
table(ldata[["ToySample"]]$tls_id_knn)
plot(ldata[["ToySample"]]$x, ldata[["ToySample"]]$y,
     col = ifelse(ldata[["ToySample"]]$tls_id_knn > 0, "red", "gray"),
     pch = 19, cex = 0.5, main = "Detected TLS in toy data")
```

plot_TLS

Plot Spatial Map of TLS and T-cell Clusters

Description

Produces a ggplot2 scatter plot of cell positions, coloured by TLS membership, T-cell cluster membership, and background phenotype.

Background (non-TLS, non-TIC) cells are rendered with a lower alpha to keep them visually recessive, while TIC cells are drawn slightly larger than TLS cells so they stand out without dominating the plot.

Usage

```
plot_TLS(
  sample,
  ldata = NULL,
  show_tic = TRUE,
  point_size = 0.5,
  alpha = 0.7,
  bg_alpha = 0.25,
  tic_size_mult = 1.8,
  tls_palette = c("#0072B2", "#009E73", "#CC79A7", "#D55E00", "#56B4E9", "#F0E442"),
  tic_colour = "#E69F00",
  bg_colour = "grey80"
)
```

Arguments

sample	Character. Sample name in ldata.
ldata	Named list of data frames, or NULL to use the global ldata object (deprecated; pass explicitly).
show_tic	Logical. Colour T-cell clusters (if detect_tic has been run) in a distinct colour? Default TRUE.
point_size	Numeric. Base point size for TLS cells and background cells (default 0.5). TIC cells are drawn at point_size * tic_size_mult.
alpha	Numeric. Point transparency for TLS and TIC cells (default 0.7).
bg_alpha	Numeric. Point transparency for background (non-TLS, non-TIC) cells (default 0.25). Reducing this value pushes background cells further behind the foreground structure.
tic_size_mult	Numeric. Multiplier applied to point_size for TIC cells so they appear slightly larger than background and TLS cells (default 1.8).

tls_palette	Character vector of colours for TLS IDs. Recycled if there are more TLS than colours. Default uses a colourblind-friendly palette.
tic_colour	Character. Colour for T-cell cluster cells (default "#E69F00").
bg_colour	Character. Colour for non-TLS, non-TIC cells (default "grey80").

Value

A ggplot object (invisibly). The plot is also printed unless the return value is assigned.

Examples

```
data(toy_ldata)
ldata <- detect_TLS("ToySample", k = 30, ldata = toy_ldata)

p <- plot_TLS("ToySample", ldata = ldata)
```

scan_clustering	<i>Scan Tissue for Local Immune Cell Clustering (Ripley's L Heatmap)</i>
-----------------	--

Description

Applies a sliding-window Ripley's L analysis across the tissue to produce a spatial clustering map. For each window a *K-integral* index is computed as the mean positive excess of the observed L function over its theoretical CSR value. When `plot = TRUE` a base-graphics spatial map is drawn with LOESS-smoothed L-excess curves and numeric CI labels overlaid inside each qualifying window, plus a legend identifying point and curve colours.

Usage

```
scan_clustering(
  ws = 500,
  sample,
  phenotype = c("T cells", "B cells", "Both"),
  plot = TRUE,
  creep = 1L,
  min_cells = 10L,
  min_phen_cells = 5L,
  label_cex = 1.1,
  ldata = NULL
)
```

Arguments

ws	Numeric. Window side length in microns (default 500).
sample	Character. Sample name in ldata.
phenotype	One of "T cells", "B cells", or "Both".
plot	Logical. Draw the spatial clustering map? (default TRUE).
creep	Integer. Grid density factor; <code>creep = 2</code> overlaps adjacent windows by half a window width, producing a smoother map (default 1).

min_cells	Integer. Minimum total cell count required in a window before it is analysed (default 10).
min_phen_cells	Integer. Minimum phenotype-specific cell count per window (default 5).
label_cex	Numeric. Base character expansion for the CI numeric labels drawn inside each window (default 1.1). Increase this value if labels appear too small for your screen or output resolution.
ldata	Named list of data frames, or NULL to use the global ldata object (deprecated; pass explicitly).

Details

The K-integral clustering index for window w is:

$$CI_w = \frac{1}{N_+} \sum_{i: L_i > L_{\text{theo},i}} (L_i - L_{\text{theo},i})$$

where N_+ is the number of spatial lags where the observed L exceeds the theoretical CSR value.

When `plot = TRUE` the map shows:

- All cells as small light-grey points.
- Phenotype cells (T cells green, B cells red).
- Navy dashed grid lines marking window boundaries.
- A LOESS-smoothed L-excess curve inside each qualifying window.
- A bold numeric CI label centred in the window.
- A legend identifying all point and curve colours.

When `phenotype = "Both"` two side-by-side panels are produced — one for B cells and one for T cells — so the two clustering maps can be compared directly on the same spatial layout.

Value

A named list with elements B and/or T (depending on phenotype), each containing the `Lest` objects for all qualifying windows of that phenotype. Returned invisibly when `plot = TRUE`.

Examples

```
data(toy_ldata)

L_models <- scan_clustering(
  ws      = 200,
  sample  = "ToySample",
  phenotype = "B cells",
  plot    = TRUE,
  ldata   = toy_ldata
)
cat("B-cell windows analysed:", length(L_models$B), "\n")

# Side-by-side B and T cell panels
L_both <- scan_clustering(
  ws      = 200,
  sample  = "ToySample",
  phenotype = "Both",
  plot    = TRUE,
```



```

      ldata      = toy_ldata
    )

```

summarize_TLS

Summarize Detected TLS Across Samples

Description

Produces a tidy data.frame with one row per sample summarising the number of detected TLS, their sizes, and (optionally) ICAT scores.

Usage

```
summarize_TLS(ldata, calc_icat_scores = FALSE)
```

Arguments

ldata Named list of data frames as returned by [detect_TLS](#) (and optionally [detect_tic](#)).

calc_icat_scores Logical. Should ICAT scores be computed for each TLS and appended as a list-column? Default FALSE.

Value

A data.frame with columns:

sample Sample name.

n_TLS Number of TLS detected.

total_cells Total cells in the sample.

TLS_cells Number of cells assigned to any TLS.

TLS_fraction Fraction of all cells that are TLS cells.

mean_TLS_size Mean cells per TLS (NA if n_TLS = 0).

n_TIC Number of T-cell clusters detected by [detect_tic](#) (NA if not yet run).

icat_scores List-column of ICAT scores per TLS (only when calc_icat_scores = TRUE).

Examples

```

data(toy_ldata)
ldata <- detect_TLS("ToySample", k = 30, ldata = toy_ldata)
summarize_TLS(ldata)

```

toy_ldata	<i>Toy Multiplexed Imaging Data</i>
-----------	-------------------------------------

Description

A small synthetic dataset mimicking multiplexed tissue imaging data, used in package examples and tests. The list contains one sample named "ToySample".

Usage

toy_ldata

Format

A named list with one element:

ToySample A data.frame with the following columns:

- x Numeric. X coordinate in microns.
- y Numeric. Y coordinate in microns.
- phenotype Character. Cell phenotype label. Values are "B cell", "T cell", and "Other".

Source

Synthetically generated for package examples.

References

Amiryousefi et al. (2025) [doi:10.1101/2025.09.21.677465](https://doi.org/10.1101/2025.09.21.677465)

Examples

```
data(toy_ldata)
str(toy_ldata)
table(toy_ldata[["ToySample"]]$phenotype)
```